

Experiment 2: Setting Up a Zoho Creator Account and Exploring the Platform Architecture and Interface

Aim:

To create a Zoho Creator account and explore the platform architecture, features, and user interface for developing low-code applications.

Theory:

Zoho Creator is a cloud-based low-code application development platform that enables users to build custom business applications with minimal coding. It provides a visual development environment where applications can be designed using drag-and-drop components, forms, workflows, reports, and dashboards.

The platform architecture consists of the following components:

- **Applications:** Containers that hold all forms, reports, workflows, pages, and settings.
- **Forms:** Used to collect and store user data.
- **Reports:** Display and manage stored data in various formats such as lists, tables, and calendars.
- **Pages:** Custom interfaces that combine multiple components into a single view.
- **Workflows:** Automate tasks using event-based actions and business logic.
- **Deluge Script:** Zoho Creator's scripting language used to implement advanced functionality and automation.
- **Users and Permissions:** Control access to applications and define user roles.

The Zoho Creator interface includes the Home Dashboard, Application Builder, Form Designer, Report Builder, Workflow Builder, Navigation Panel, and Settings menu, allowing users to efficiently design, test, and deploy applications.

Objectives:

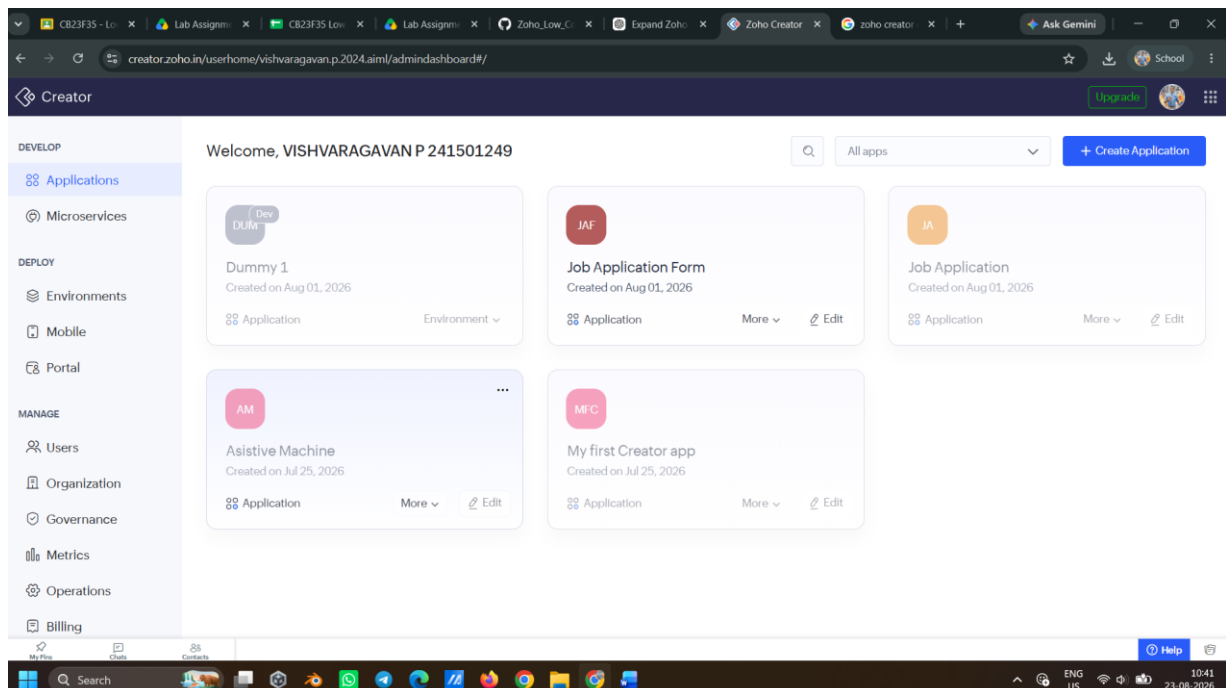
Add a separate **Objectives** section:

- To create and configure a Zoho Creator account.
- To understand the concept of low-code application development.
- To explore the Zoho Creator platform architecture.
- To understand applications, forms, reports, pages, and workflows.
- To explore the Zoho Creator user interface.
- To understand data collection and storage through forms.
- To understand different methods of displaying data.
- To explore workflow automation.
- To understand Deluge scripting.
- To examine user roles and permissions.
- To understand the basic application development lifecycle.

1. Introduction to Zoho Creator

Zoho Creator is a cloud-based low-code application development platform used to create customized business applications. It provides visual tools such as forms, reports, pages, workflows, and dashboards, allowing applications to be developed with minimal programming.

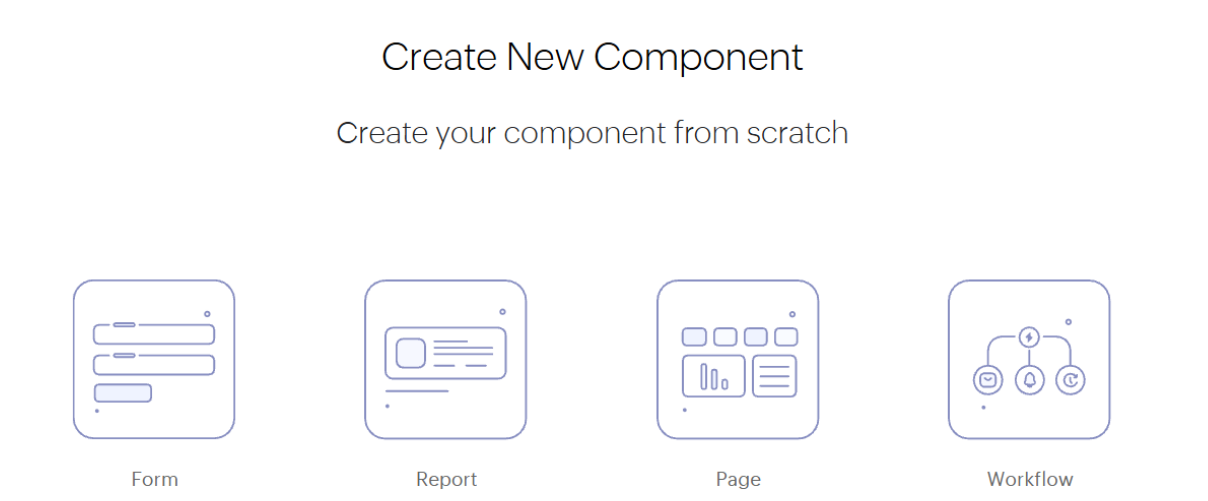
Image: Zoho Creator homepage/dashboard **Caption:** Figure 2.1: Zoho Creator Platform Homepage



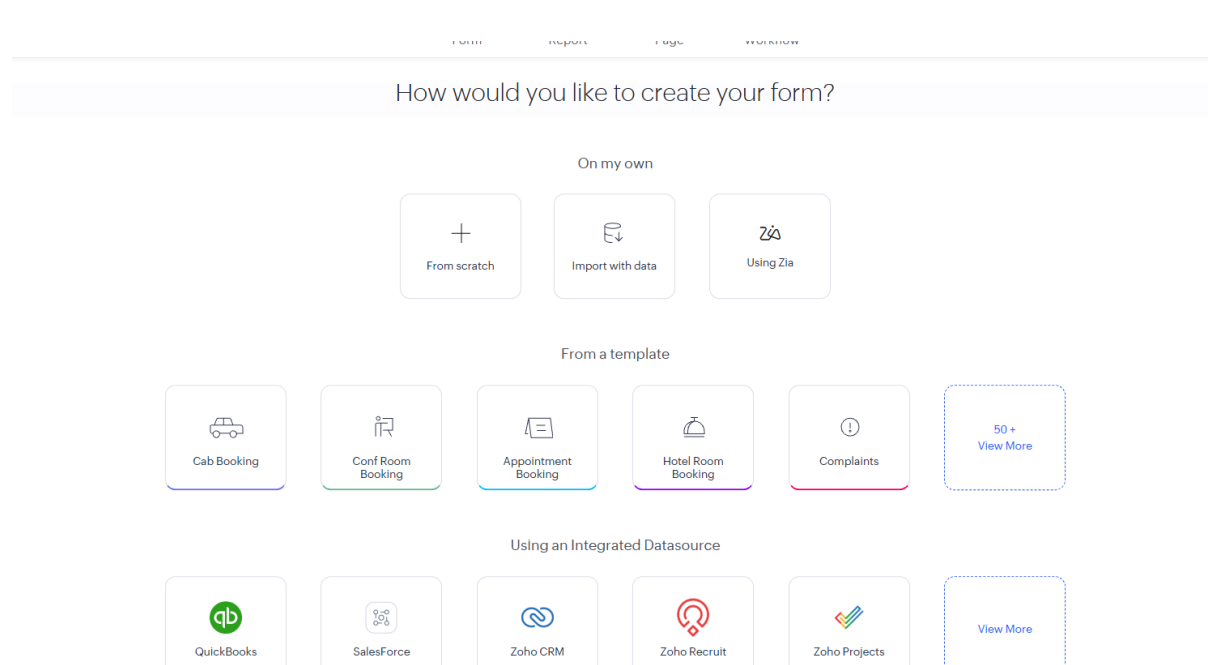
2. Low-Code Development

Low-code development allows applications to be created using visual interfaces, drag-and-drop components, and predefined functions instead of writing large amounts of traditional code. Zoho Creator follows this approach to reduce development time and simplify application development.

Image: Zoho Creator Application Builder



Caption: Figure 2.2(1): Low-Code Application Development Interface

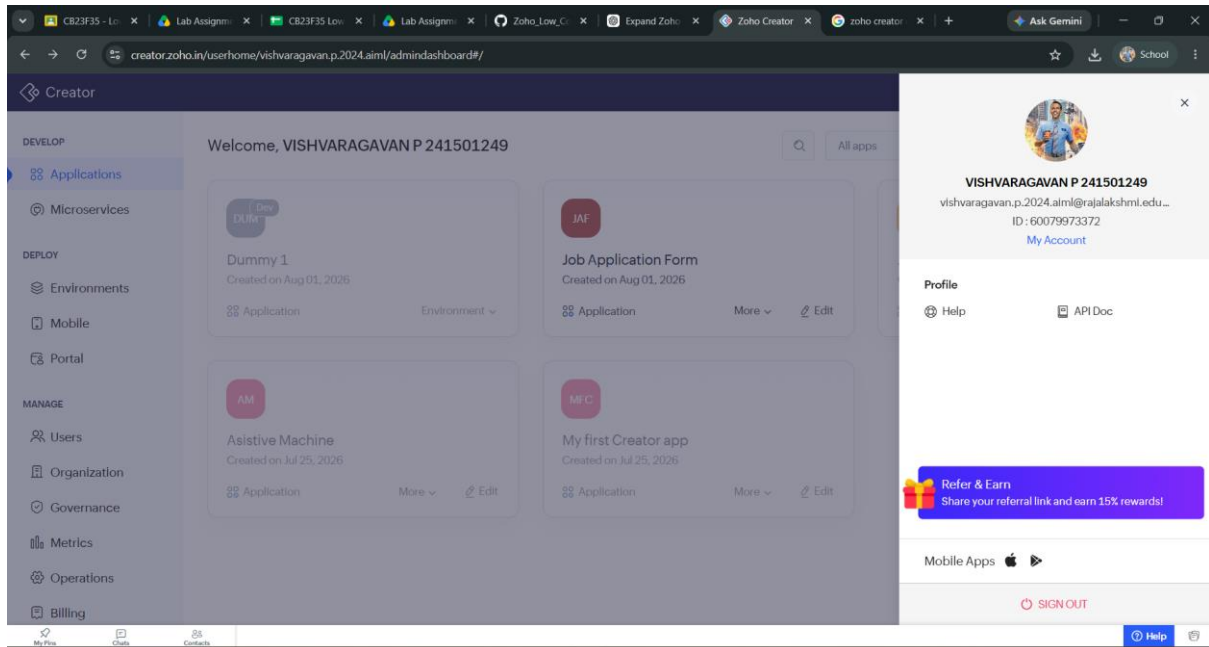


Caption: Figure 2.2(2): Low-Code Application Development Interface

3. Zoho Creator Dashboard

The Zoho Creator dashboard provides the main entry point for managing applications and accessing different platform features. It allows users to create new applications, open existing applications, and manage development activities from a centralized interface.

Image: Your Creator dashboard



Caption: Figure 2.3: Zoho Creator Dashboard

4. Creating an Application

In Zoho Creator, an application functions as a self-contained digital ecosystem that consolidates all the necessary elements to automate a business process or manage specific data.

Core Components

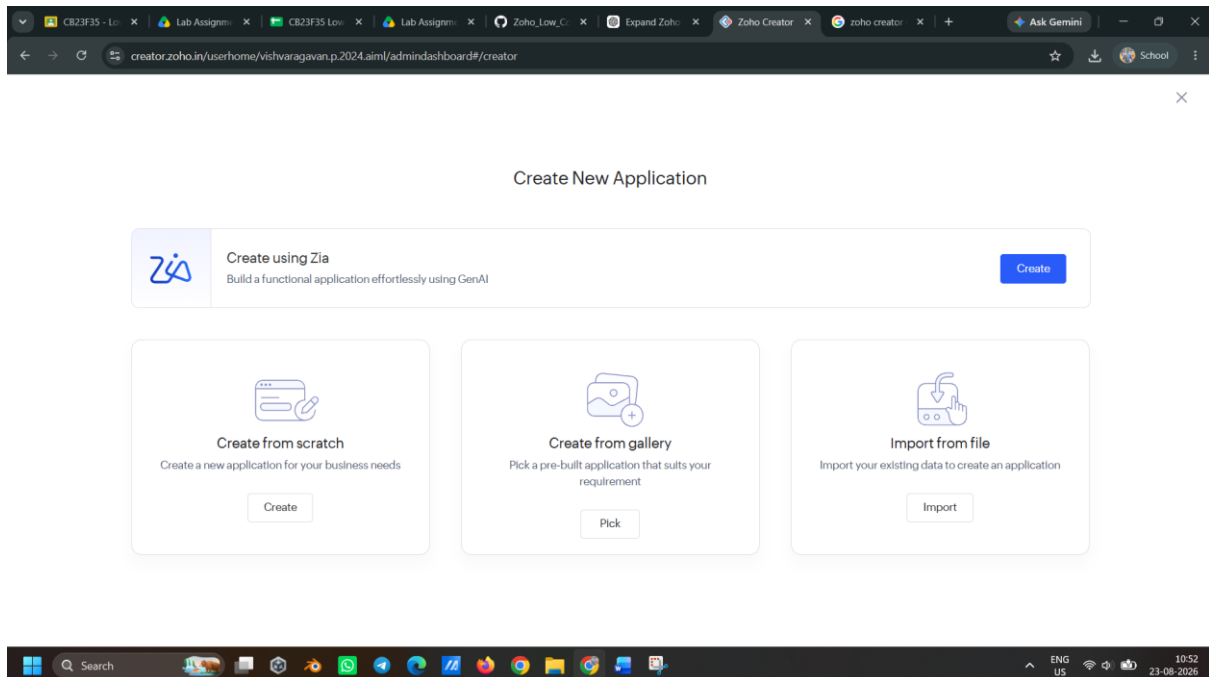
- **Forms:** Collect and capture data from users, serving as the entry point for all information.
- **Reports:** Display, filter, analyze, and export stored data using dynamic visual formats like grids, Kanban boards, charts, and maps.
- **Pages:** Custom dashboards built using visual elements and widgets to give users an interactive interface for viewing metrics and navigating the app.
- **Workflows:** Automate repetitive business logic, notifications, integrations, and approvals using visual builders or Zoho's proprietary Deluge scripting language.
- **Settings:** Control global configurations, such as user roles, access permissions, branding, and third-party API connections.

Creation Methods

- **From Scratch:** Build completely custom applications starting with a blank canvas to match unique, highly specific operational workflows.

- **From Templates:** Launch apps instantly using pre-built, industry-specific solutions that can be modified to accelerate development time.

Image: Create Application screen



Caption: Figure 2.4: Creating a New Application

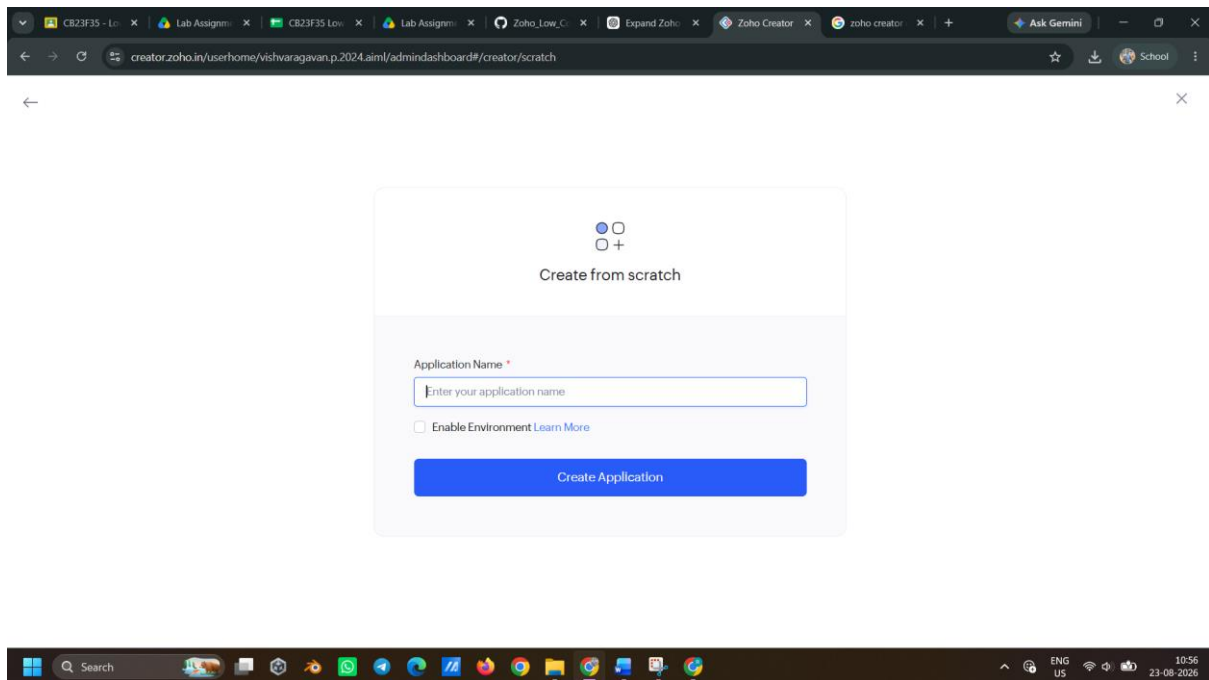
5. Application Builder

Application Life Cycle Management (ALM) in Zoho Creator provides the environment to manage an application's entire journey from initial development to deployment and maintenance.

Core Aspects of ALM

- **Multi-Environment Staging:** Develop and test applications in isolated Development and Sandbox environments to experiment safely without disrupting live business operations.
- **Version Control & History:** Track application changes over time, review revision logs, and restore previous versions if unexpected errors occur during updates.
- **Seamless Deployment:** Publish updates from development to production with a single click, ensuring zero downtime for active users.
- **Access & Governance:** Assign fine-grained roles and permissions to developers, admins, and end-users to secure sensitive data across environments.
- **Maintenance & Analytics:** Monitor system performance, track audit logs, manage API usage limits, and debug scripts via detailed execution logs.

Image: Application Builder interface

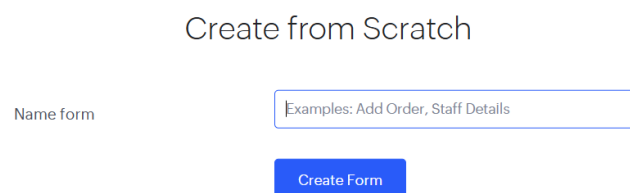


Caption: *Figure 2.5: Zoho Creator Application Builder*

6. Forms

Forms are used to collect information from users and store it as records in the application. Different fields such as text, number, email, date, dropdown, checkbox, and file upload can be added according to the information required.

Image: Form Builder with different fields



Caption: *Figure 2.6(1): Form Design Interface*

Job Application Form
Job_Application_Form

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

Number
123

Date
15

Time

Drop Down

Section A — Personal Details

Candidate_Name *

Current Address

Contact Number *

Field Properties

Field name

Candidate_Name

Field link name

Candidate_Name

[View Field Reference](#)

Validation

☒ Mandatory

Display Fields

☐ Prefix

☒ First Name

☒ Last Name

☐ Suffix

Section B — Contact & Online Presence

Email Address *

LinkedIn / Portfolio URL

Section C — Job & Experience Details

Department Applied For *

Employment Type Preferred *

Caption: *Figure 2.26: Form Design Interface*

7. Creating Form Fields

Zoho Creator provides several field types for collecting different kinds of data. During form creation, appropriate fields can be selected and configured with properties such as field name, data type, required status, and validation rules.

Image: Field selection pane

Basic Fields



Name



Email



Address



Phone



Single Line



Multi Line

123

Number



Date



Time



Drop Down

Section A — Personal Details



Candidate_Name *



Current Address



Contact Number *

Section B — Contact & Online Presence



Email Address *



LinkedIn / Portfolio URL

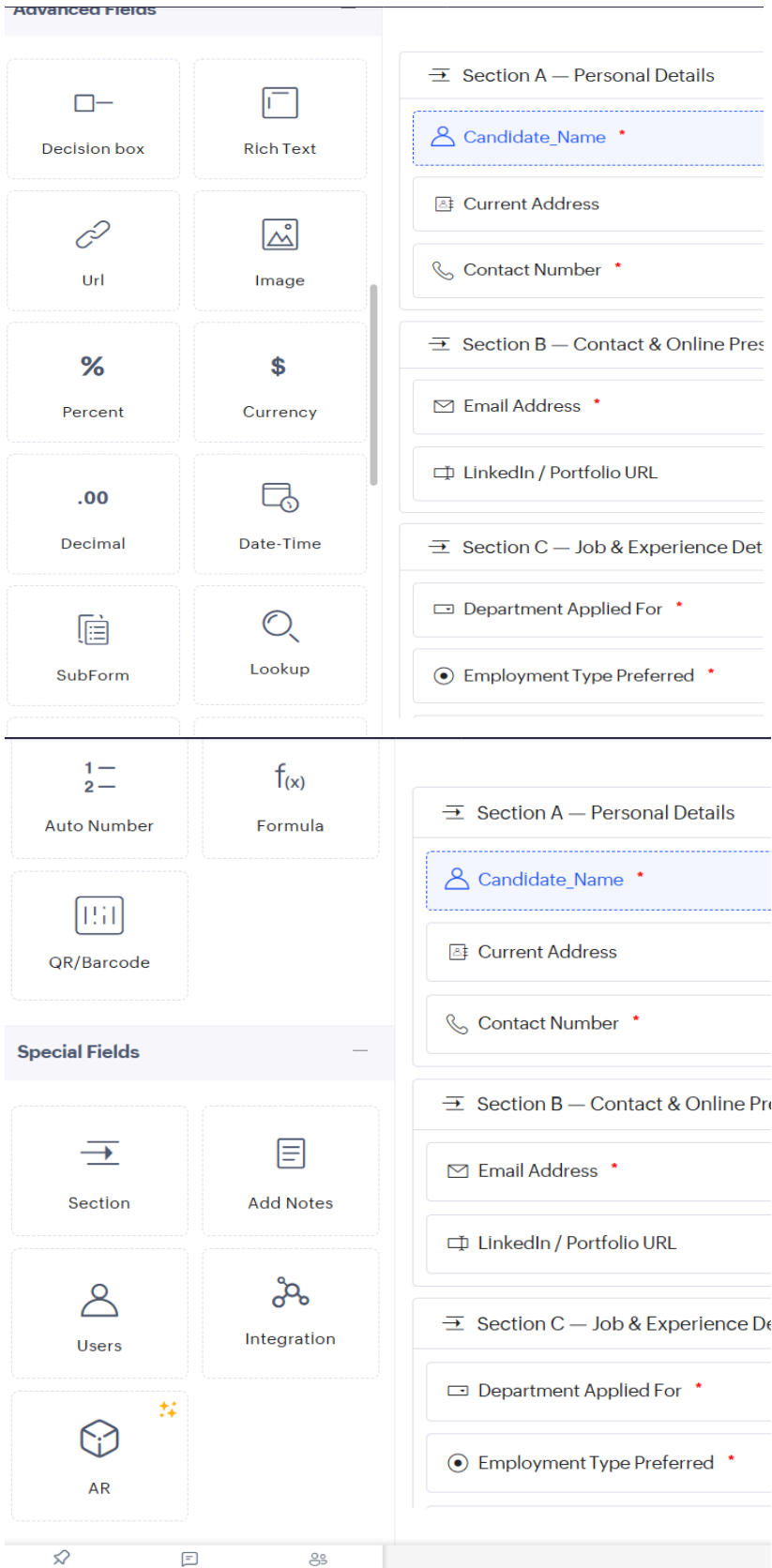
Section C — Job & Experience Details



Department Applied For *

☒

Employment Type Preferred *



Caption: Figure 2.7: Available Form Field Types

8. Data Entry

Once a form is built, Zoho Creator automatically generates a structured, mobile-responsive data-entry interface for end users. When a user submits the form, validation rules check the input before the system converts the information into a discrete record stored securely in the central database. After storage, authorized users can search, view, edit, or delete these records. These captured entries also feed directly into dynamic reports, visual dashboards, and automated workflows to drive real-time business processes.

Image: Filled sample form:

The screenshot shows a web browser displaying a 'Job Application Form' in Zoho Creator. The form is titled 'Job_Application_Form' and is filled with sample data. The left sidebar shows a user profile for 'Hello VISHVARAGAV...' and a 'Job_Application_Form' icon. The form fields are as follows:

- City / District:** NewDilli
- State / Province:** Delhi
- Postal Code:** 129900
- Country:** India
- Contact Number:** +91 32118563444
- Section B – Contact & Online Presence:**
 - Email Address:** ramnohan@gmail.com
 - LinkedIn / Portfolio URL:** https://in.linkedin.com/ramcharean
- Section C – Job & Experience Details:**
 - Department Applied For:** -Select-
 - Employment Type Preferred:**
 - ☐ Choice 1
 - ☐ Choice 2
 - ☐ Choice 3

Caption: Figure 2.8: Sample Data Entry Form

9. Records and Database

Every form submission generates an individual data record that is stored securely within Zoho Creator's relational database backend. These stored records collectively constitute the application's core dataset, providing the foundational information used across all other app modules. Once captured,

This raw data can be retrieved, filtered, and displayed through custom reports, manipulated via Deluge script workflows

Embedded into dashboard pages, or shared across external applications through webhooks and APIs.w

Image: Filled sample form

Candidate Name	Email Address	Current Address	Contact Number	LinkedIn / Portfolio	Cover Letter
Rohan Sharma	rohan.sharma@email.com	42, Park Avenue, Indiranagar, Bengaluru, Karnataka, 560038, India	+919876543210	Rohan/linkedin.com	Dear Hiring Team, I am an Engineering position : developing robust app ability to add immedi Show more

Caption: Figure 2.9: Stored Application Records

10. Reports

Reports are used to display and manage the records stored in an application. Zoho Creator provides different report formats that help users view information in an organized and meaningful manner.

Image: Report Builder

Form

Report

Page

Workflow

Which type of report would you like to make?

List

Pivot Chart

Pivot Table

Spreadsheet

Kanban

Calendar

Timeline

Map

Caption: Figure 2.10: Report Builder Interface

11. Table/List Report

Table and list reports display application records in an organized format. They allow users to view multiple records at once and perform operations such as searching, filtering, sorting, editing, and deleting records based on their permissions.

Image: Table/List report

Report as a List

Based on the form

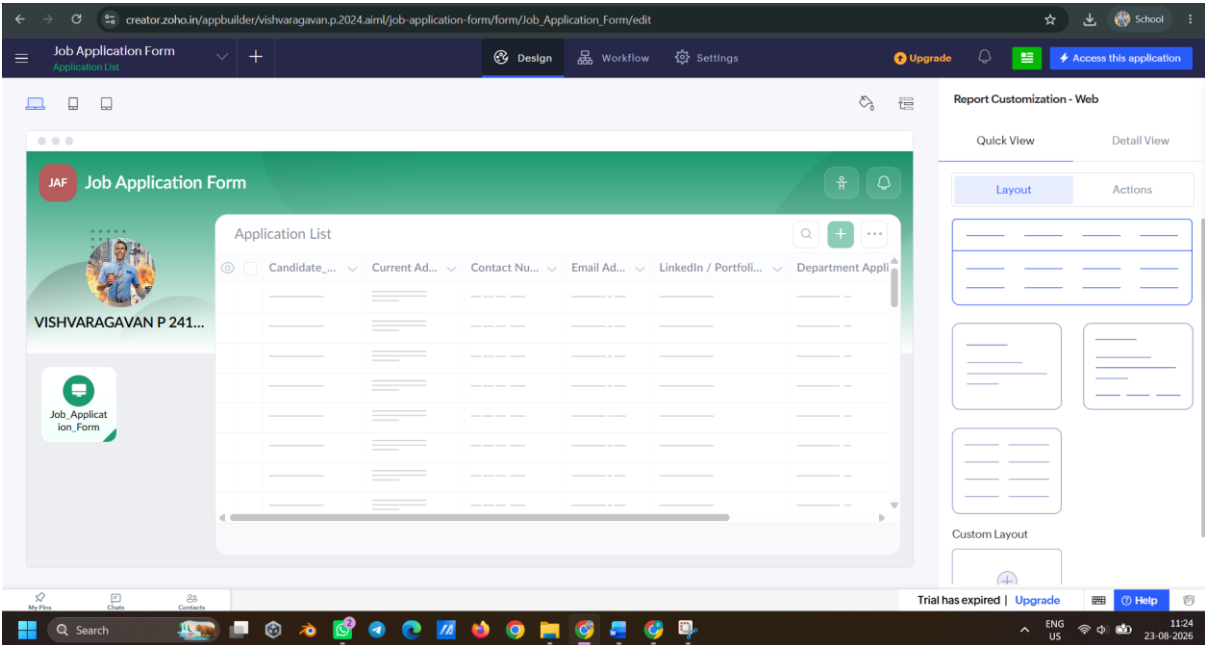
Job_Application_Form

Name report

Application List

Create Report

Caption: Figure 2.11(1): Student Records Report



Caption: Figure 2.11(2): Student Records Report

12. Chart Report

Chart reports provide a graphical representation of stored data. Charts make it easier to identify patterns, compare categories, and understand large amounts of information quickly.

Image: Bar chart, pie chart, or other chart

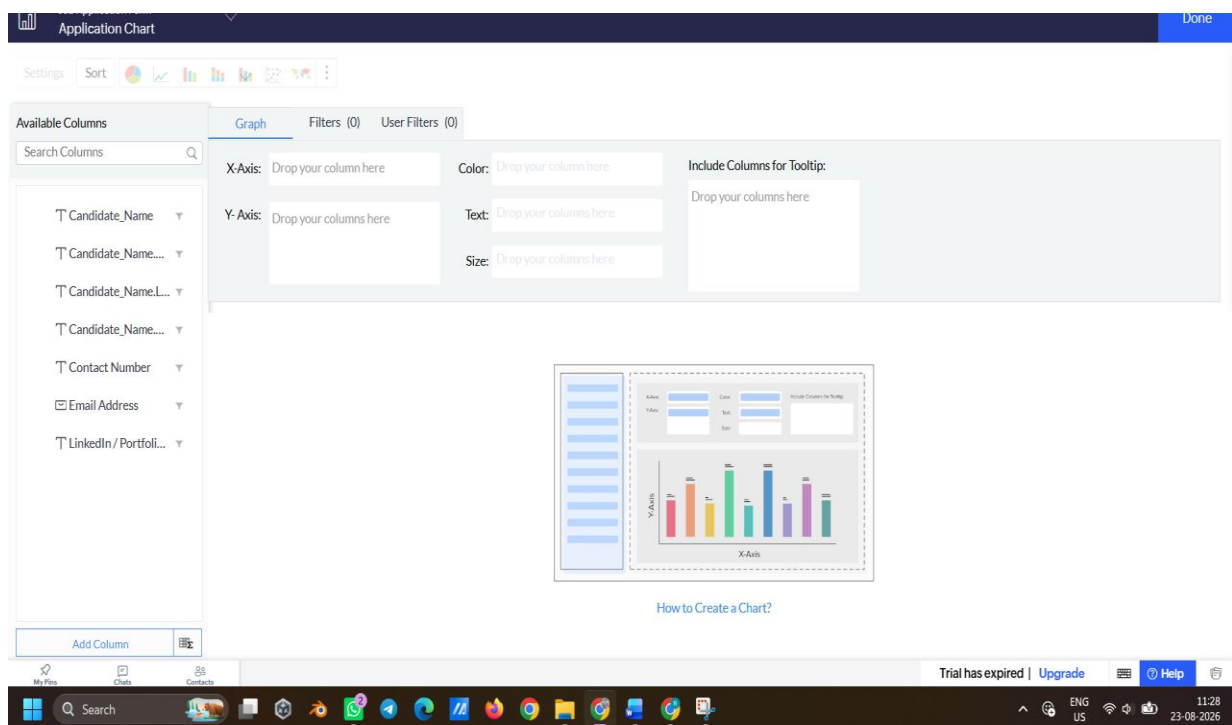
Report as a Pivot Chart

Based on the form Job_Application_Form

Name report Application Chart

[Creating](#)

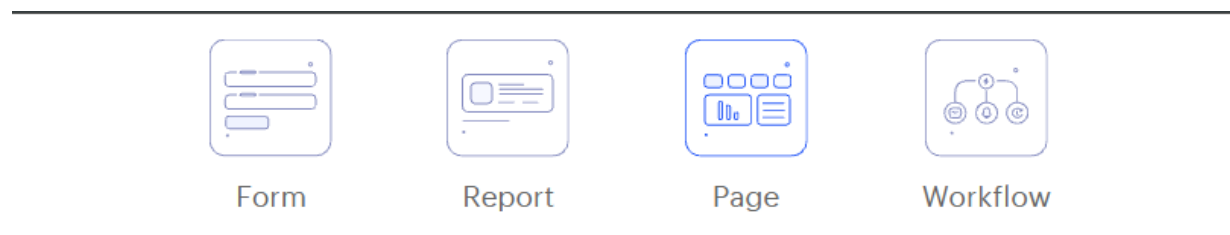
Caption: Figure 2.12(1): Graphical Data Representation



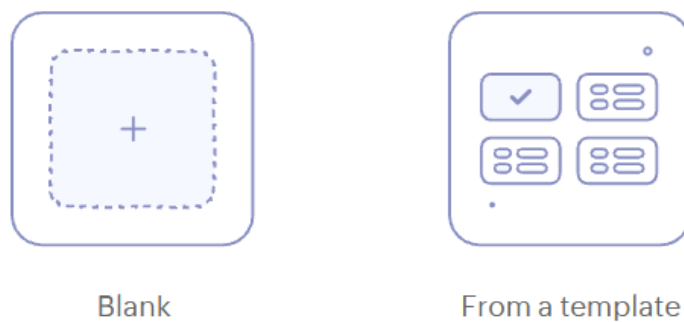
Caption: Figure 2.12(2): Graphical Data Representation

Pages are customizable interfaces used to combine different application components into a single screen. They can contain reports, charts, buttons, images, text, and other elements to create user-friendly application dashboards.

Image: Page Builder



How would you like to create your page?



Caption: Figure 2.13(1): Zoho Creator Page Builder

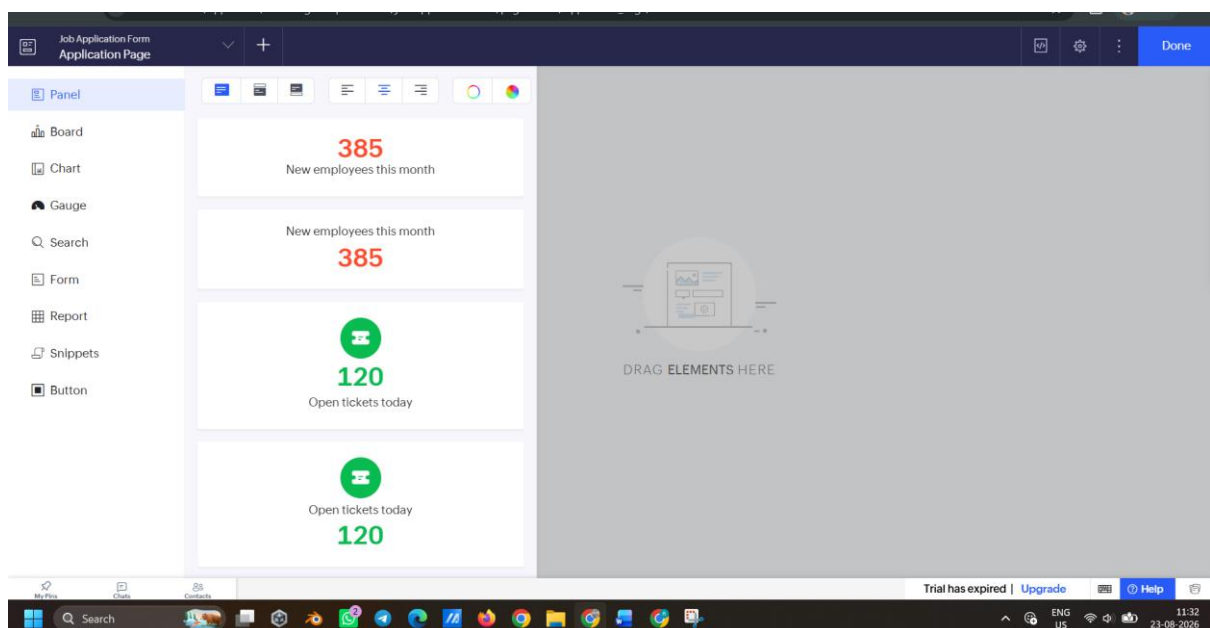
Create New Page

Name page

Application Page

Create Page

Caption: Figure 2.13(2): Zoho Creator Page Builder



Caption: Figure 2.13(3): Zoho Creator Page Builder

14. Dashboard

A dashboard provides a visual summary of important application information. Multiple components such as charts, reports, counters, and tables can be placed on a dashboard to help users monitor data and make decisions efficiently.

Image: Completed dashboard

Caption: Figure 2.14: Application Dashboard

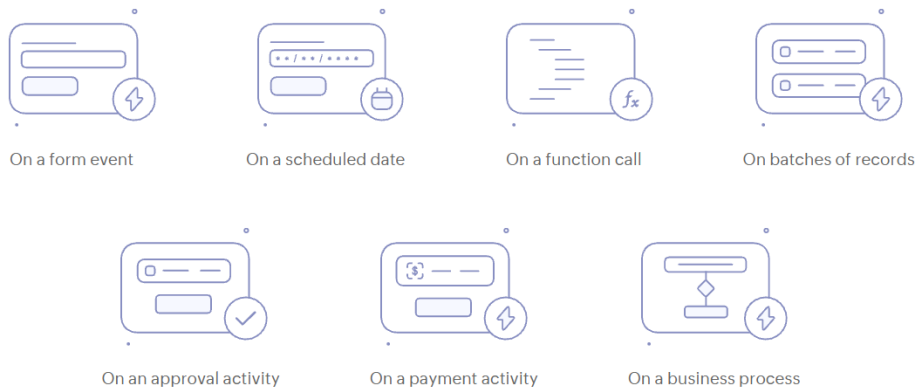
15. Workflows

Workflows serve as the automated engine of a Zoho Creator application, executing preset business logic when specific actions or conditions occur. They can be triggered by form events (such as adding, editing, or deleting a record), timed schedules, or custom user buttons.

When an event takes place—for example, a new student record being submitted—the workflow evaluates conditional logic to run automated tasks automatically. It can validate incoming data to prevent errors, perform real-time calculations to update related fields across the database, and trigger instant multi-channel notifications via email, SMS, or system alerts. Workflows also handle complex business processes like routing approval requests to managers, calling external webhooks, and executing custom Deluge scripts to sync data with third-party applications seamlessly.

Image: Workflow Builder:

When do you want to run your action?



Caption: *Figure 2.15: Workflow Automation*

Run workflow on any event in the form

Choose form

Job_Application_Form

Run when a record is
(Record Event)

☐ Created

☐ Edited

☐ Created or Edited

☐ Deleted

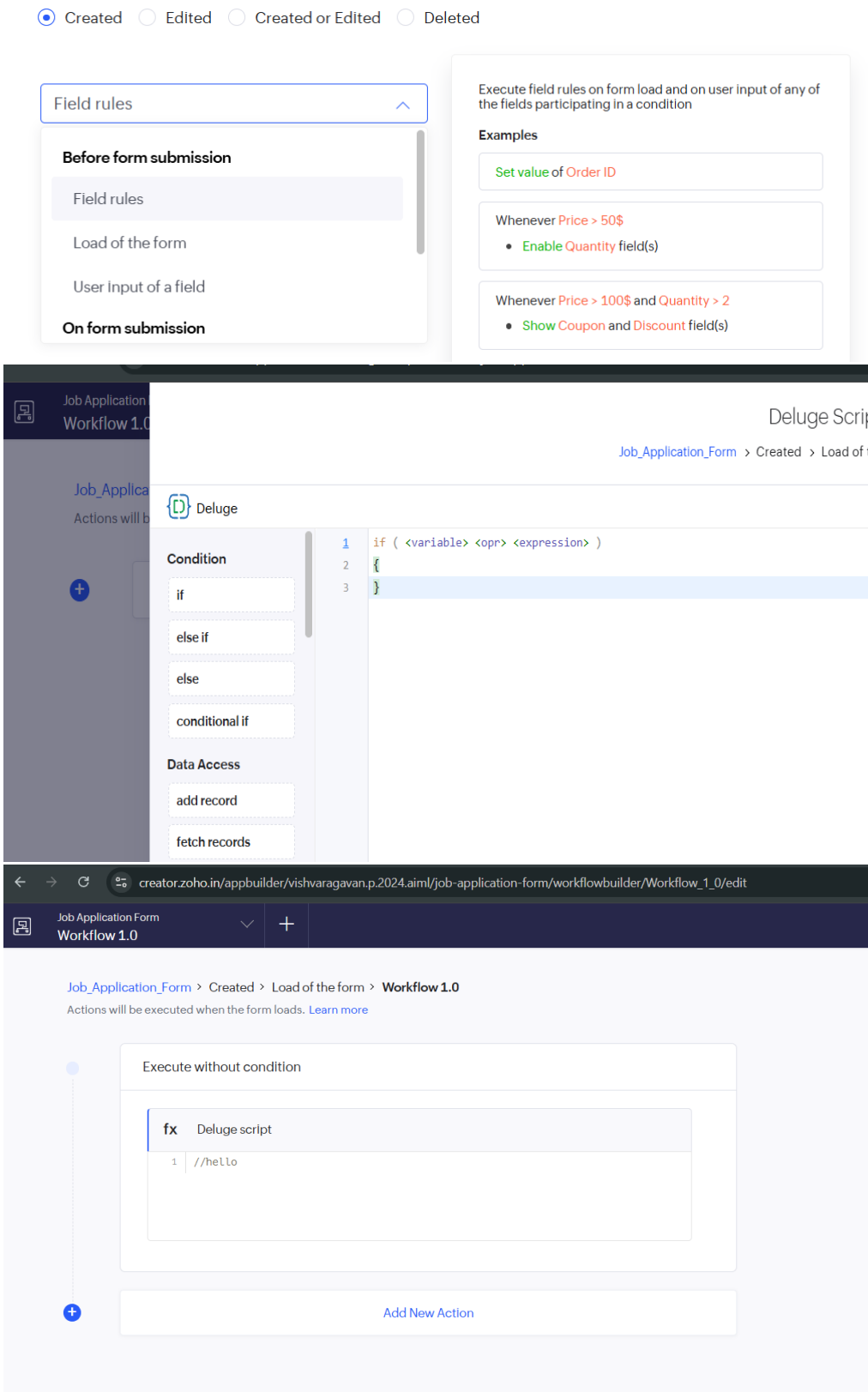
When to trigger workflow
(Form Event)

-Select-

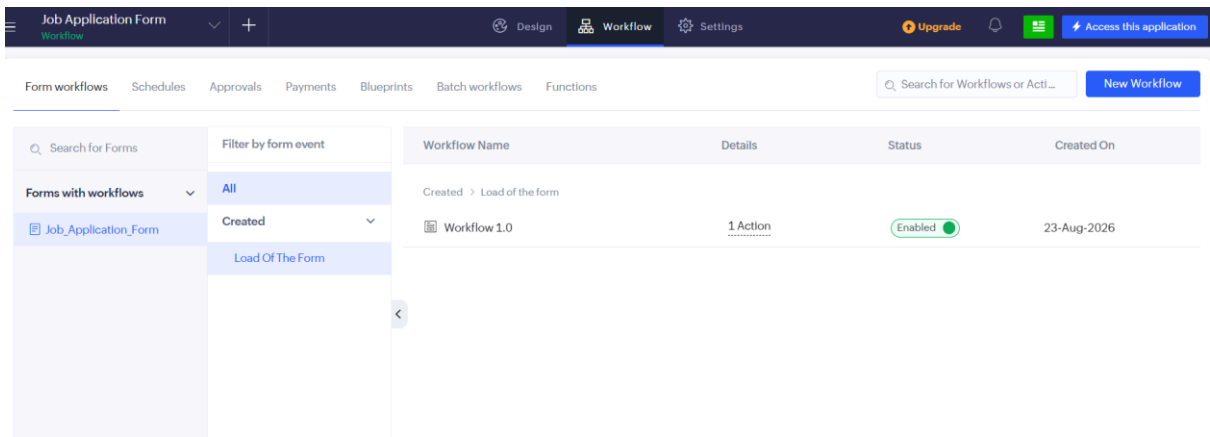
Name the workflow

Examples: Notify Customer, Calculate Sum

Create Workflow



Caption: Figure 2.15: Workflow Automation

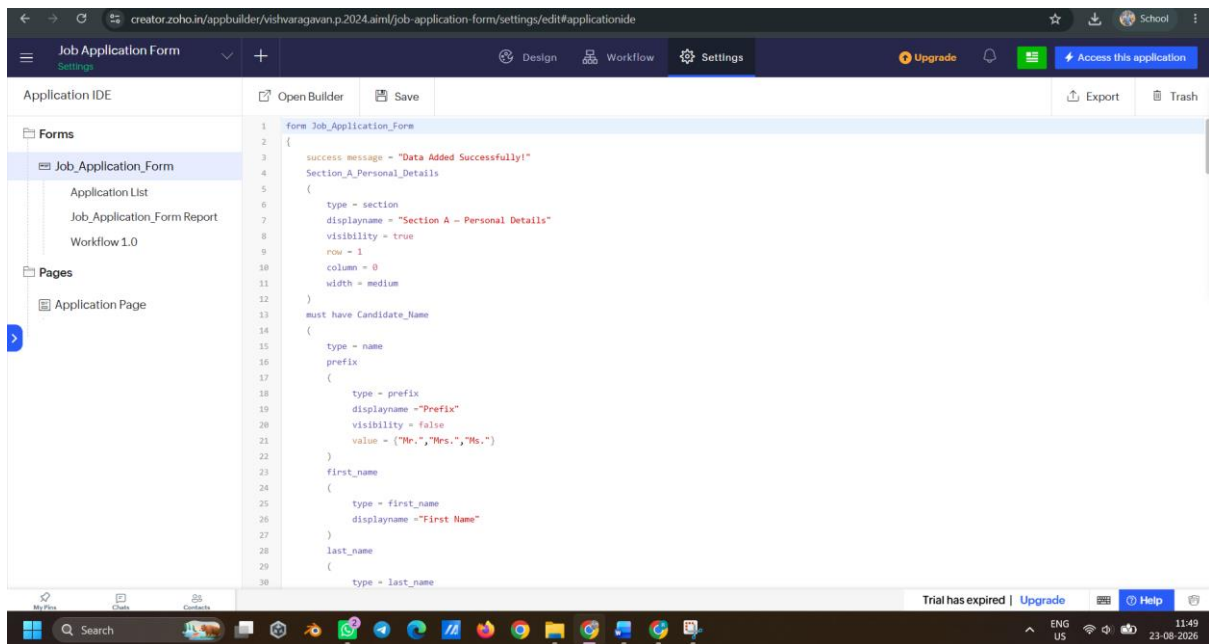


16. Deluge Scripting

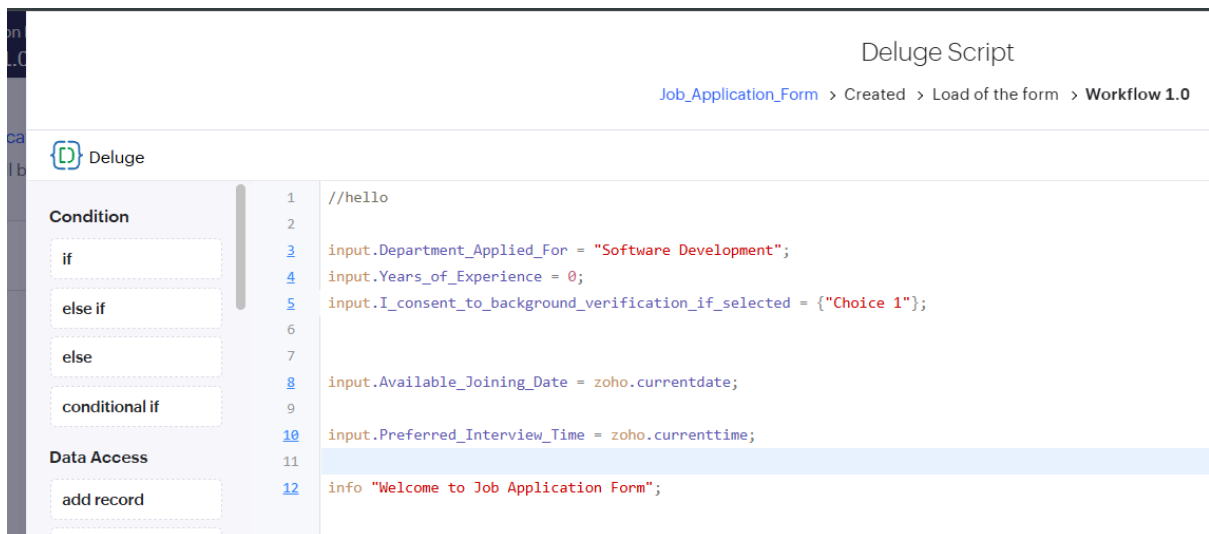
Deluge (Data Enriched Language for Universal Grid Environment) is Zoho's proprietary low-code scripting language, designed specifically to add deep custom logic and complex processing capabilities to your application. It acts as the functional backbone for tailoring app behavior beyond standard drag-and-drop settings.

- **Calculations:** Performs real-time mathematical operations, string manipulations, and field value transformations dynamically upon data entry or updates.
- **Data Validation:** Enforces custom business rules and constraints before records are saved, preventing duplicate entries or invalid inputs.
- **Conditional Operations:** Executes targeted actions based on precise logical conditions, branching workflows depending on user inputs or system states.
- **Record Manipulation:** Programmatically creates, fetches, updates, or deletes records across multiple forms and tables within the application or between connected Zoho apps.
- **Advanced Workflow Automation:** Connects external services via APIs and webhooks, routes multi-level approval processes, and drives complex business routines seamlessly.

Image: Deluge script editor:



Also we can see :



Caption: Figure 2.17: **Deluge Script Editor**

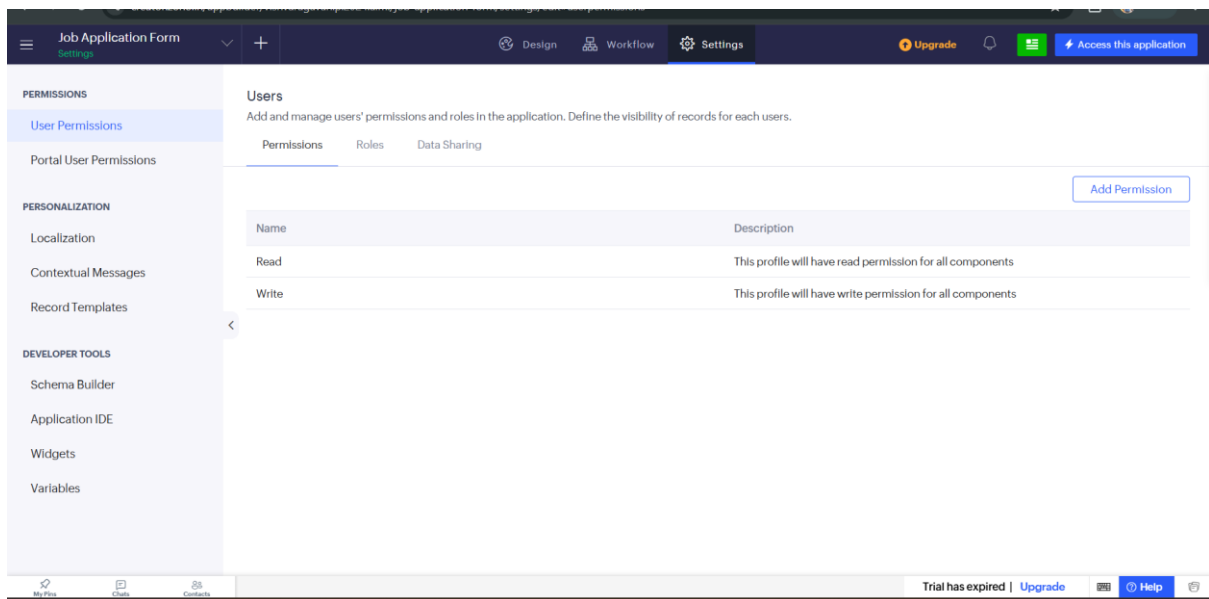
17. Permissions

Permissions define the exact level of access and operational authority assigned to different users within a Zoho Creator application. By establishing security boundaries

Permissions ensure users can only see and interact with the data necessary for their specific roles, protecting sensitive information and maintaining data integrity.

- **Role-Based Access:** Categorizes users by organizational hierarchy (such as Admins, Managers, or Field Staff) to grant appropriate administrative, operational, or view-only rights.
- **Component-Level Control:** Restricts or enables access to specific forms, reports, pages, and dynamic widgets based on user profiles.
- **Record-Level Rights:** Dictates precisely what a user can do with individual data entries, determining whether they can create new records, edit existing ones, delete entries, or export reports.
- **Field-Level Security:** Allows administrators to hide or lock specific fields within a form—such as salary details or internal notes—so they remain hidden from unauthorized users while staying visible to managers.

Image: Permission configuration



Caption: *Figure 2.19: Application Permissions*

Procedure:

The experiment was performed by creating and exploring a Zoho Creator application step by step. The process included setting up the account, creating an application, designing forms, adding fields, entering sample records, generating reports and charts, creating pages and dashboards, configuring workflows, exploring Deluge scripting, managing users and permissions, and finally testing the complete application. Screenshots were captured at important stages to document the development process and understand the overall platform architecture.

1. Open a web browser and navigate to the **Zoho Creator** website.
2. Create a new Zoho account using the required credentials, or sign in using an existing Zoho account.
3. Open **Zoho Creator** from the Zoho application dashboard and access the Creator home page.
4. Explore the **Zoho Creator Dashboard** and identify the available options for creating, opening, and managing applications.
5. Create a new application by selecting either a **blank application** or a suitable **pre-built template**.
6. Enter an appropriate application name and open the newly created application in the **Application Builder**.
7. Explore the **Application Builder interface** and identify the main sections used for designing and managing the application.
8. Create a new **Form** inside the application for collecting and storing user information.
9. Add different field types to the form, such as **Single Line, Number, Email, Date, Dropdown, Radio Button, Checkbox, and File Upload**, according to the requirements.
10. Configure the properties of the fields by setting appropriate field names, data types, mandatory fields, and validation options.
11. Save the form and open the generated **data-entry interface**.
12. Enter sample information into the form and submit the form to create application records.
13. Open the stored records and verify that the submitted information has been successfully saved.
14. Explore the available options for **searching, filtering, sorting, editing, and deleting records**.
15. Create a new **Report** based on the form data to display the stored records in an organized format.
16. Explore different report formats such as **list, table, summary, chart, and calendar views**, wherever available.

17. Create a **table or list report** and verify that the submitted records are displayed correctly.
18. Create a **chart report** using the stored data and observe how the information can be represented graphically.
19. Create a new **Page** and explore the Page Builder interface.
20. Add suitable components such as **reports, charts, buttons, text, images, and other widgets** to the page.
21. Arrange the components to create a simple **application dashboard** for displaying important information.
22. Open the **Workflow** section and create a sample workflow based on a suitable form event.
23. Configure a workflow trigger such as **adding, editing, or deleting a record**.
24. Add conditions and actions to the workflow to demonstrate basic automation.
25. Test the workflow by performing the event that triggers it and verify that the configured action is executed successfully.
26. Open the **Deluge Script** section and explore the scripting environment provided by Zoho Creator.
27. Create or examine a simple Deluge script for performing an operation such as **calculation, validation, conditional processing, or record manipulation**.
28. Execute and test the script and observe the result produced by the application.
29. Explore the **Users and Permissions** section of the application.
30. Examine the available user roles and configure suitable permissions for different types of users, such as administrators and normal users.
31. Verify that users with different permissions can perform only the operations allowed for their assigned roles.
32. Open the **Application Settings** section and explore the available configuration options related to the application.
33. Examine the application navigation structure and observe how users can move between **forms, reports, pages, and dashboards**.
34. Test the complete application by entering valid and invalid sample data and checking whether the configured validation rules work correctly.
35. Test the major application functions, including **record creation, viewing, editing, deletion, report generation, dashboard display, workflow execution, and permissions**.
36. Observe how the different components are connected, following the basic flow:
User → Form → Record/Data → Report/Page → Workflow → Output
37. Capture screenshots of the important stages, including the **Zoho Creator dashboard, application creation, Application Builder, Form Builder, fields,**

submitted records, reports, charts, pages, dashboard, workflow, Deluge editor, and permissions.

38. Analyze the role of each component and understand how Zoho Creator integrates forms, data storage, reports, pages, workflows, scripting, and user permissions into a single low-code application platform.
39. Finally, review the completed application and verify that all major components function correctly.

Result:

A Zoho Creator account was successfully created, and the platform architecture and user interface were explored. The experiment demonstrated that Zoho Creator provides an intuitive low-code environment with tools such as forms, reports, workflows, pages, and dashboards, enabling rapid application development with minimal programming effort.